

- Q.28 Write a Note on the Lancashire boiler.
- Q.29 What is difference between evaporator and condenser?
- Q.30 What is boiler. Write a note on any one boiler
- Q.31 Describe multiple effect evaporation and its advantage in industrial application.
- Q.32 Explain the working of plate type heat exchanger
- Q.33 Discuss about the muffle furnace.
- Q.34 Describe the working of contact condenser in detail
- Q.35 What are the advantage and disadvantage of Babcock and Wilcox boilers?

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Classify different types of heat exchangers. Explain plate type heat exchanger with neat and clean diagram
- Q.37 Explain the following in detail with diagram
- Long Tube Evaporator
 - Muffle Furnace
- Q.38 Define evaporator. Explain construction and working of long tube evaporator with neat sketch, advantages and disadvantages.

No. of Printed Pages : 4

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Roll No.

4th Sem / Chem, P & P Subject:- Heat Transfer - II

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following is not a type of boiler?
- Lancashire boiler
 - Babcock and Wilcox boiler
 - Shell and tube boiler
 - Nestler boiler
- Q.2 As the Temperature increase, thermal conductivity of Gas
- Increase
 - Decrease
 - remain same
 - None
- Q.3 Which of the following is a method of heat transfer?
- Convection
 - Radiation
 - Conduction
 - All of the mentioned
- Q.4 Which of the following is the rate of heat transfer unit?
- Watt
 - Pascal
 - Joule
 - Newton

- Q.5 In liquids and gases, heat transmission is primarily caused by
- Convection
 - Radiation
 - Conduction
 - None
- Q.6 Boiling refers to a change from the
- Solid to a liquid phase
 - Vapor to a liquid phase
 - Liquid to a solid phase
 - Liquid to a vapour phase
- Q.7 Which type of boiling occurs in steam boilers employing natural convection?
- Forced convection
 - Pool
 - Local
 - Saturated
- Q.8 LMTD in case of counter flow as compared to parallel flow heat exchanger is
- Higher
 - Lower
 - remain same
 - None of these
- Q.9 The boiling phenomenon is known to occur in how many forms?
- 1
 - 2
 - 3
 - 4
- Q.10 Joule sec is the unit of
- Work Transfer
 - Planck's constant
 - Universal gas constant
 - None

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Name a process in which heat transfer is accompanied by phase change.
- Q.12 Give one application of double pipe heat exchanger.
- Q.13 Define the evaporation capacity.
- Q.14 Write the SI unit of heat transfer coefficient.
- Q.15 Name any one type of fins.
- Q.16 Write full form of LMTD.
- Q.17 Write one advantage of Lankashire boiler.
- Q.18 Smooth surface is required for _____ condensation.
- Q.19 Define overall heat transfer coefficient.
- Q.20 Write unit of fouling factor.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Describe the working of open pan evaporator.
- Q.22 Explain the working of shell and tube condenser.
- Q.23 Explain the co-current flow arrangement in a heat exchanger with the help of a neat diagram. Also write an expression for calculation of LMTD for it.
- Q.24 Discuss concept of elevation in boiling point in brief.
- Q.25 Explain the difference between dropwise and filmwise condensation.
- Q.26 Compare co-current and counter-current flow in heat Exchangers.
- Q.27 Draw the neat clean diagram of contact condenser.

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